- Sohyla Said Mohamed
- 1202006 1
- CSE - Sec 1

1.4) a- $V_{iH} = V_i + (9.81 - 0.25168 \times V_i^2)$, $V_0 = 0$

t(s)	(m/s) V
0	0
1	9.81
2	19.267
3	27.714
4	34.044
5	40.099
6	44.091
7	46.7041
8	48.5065
9	49.69
10	50.433
11	50.903
12	51.203
∞	51.4

b) $v_{i+} = v_i + \left(9.8 - \frac{0.25}{68.1} v_i^2\right) \times 0.5$

t(s)	v(m/s)	t(s)	v(m/s)
0	0	6.5	44.52
0.5	4.905	7	45.73
1	9.766	7.5	46.84
1.5	14.5	8	47.72
2	19.02	8.5	48.44
2.5	23.26	9	49.04
3	27.17	9.5	49.53
3.5	30.72	10	49.93
4	33.69	10.5	50.26
4.5	36.69	11	50.53
5	39.12	11.5	50.75
5.5	41.2	12	50.93
6	43.001	∞	≈ 51.7

* As we known that the accuracy increase when step size decrease

* and also observe this in the previous example The accuracy when using 0.5 step much better than using 1